Algebra of events

Outcomes, sample space, events

In probability theory, we make mathematical models to describe chance experiments. For a chance experiment, we identify a set of possible outcomes, which we call the sample space Ω for the experiment.

A good example to keep in mind is the chance experiment where we roll two dice, a red die and a green die. Durrett gives a sample space Ω to model this experiment in Example 1.2. He describes outcomes as ordered pairs like (1, 2) and (4, 6). The first number in the ordered pair is the number we roll on the red die, and the second number is the number we roll on the green die.

When we think about a chance experiment, we are often interested not just in specific outcomes but also in more complex **events**. An event is a collection of outcomes. In more technical mathematical terminology, we say that an event is a subset of the sample space Ω. In Durrett's dice-rolling situation, examples of events include "the red roll is bigger than the green roll" (with 15 outcomes) and "the red roll and the green roll differ by at most 1" (with 16 outcomes).

Algebra of events

In a mathematical model of a chance experiment, we assign probabilities (real numbers between 0 and 1) to events. Before we think about probabilities, though, we recognize an "algebra" of events. The following table describes the algebraic operations for events and shows some rough analogies with algebra for numbers.

Algebra of numbers	Algebra of events
0	$\varnothing$: This event is called the empty set or the empty event. It is the event that has no outcomes.
1	Ω: This is the event that includes all possible outcomes.
$x + y$	$A \cup B$: This event is called the union of A and B. It includes all outcomes in A or in B or in both.
$x \cdot y$	$A \cap B$: This event is called the intersection of A and B. It includes all outcomes that are both in A and in B.
$-x$	A^c: This event is called the complement of A. It includes all the outcomes in Ω that are not in A.

The kind of algebra that we do with events comes up many places in mathematics and computer science. Sometimes it is called “Boolean algebra.”

Basic rules of algebra of events

You don’t have to memorize the rules of algebra with events, but it can help to see them written down. The following table shows the basic rules.

$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$	associative laws
$A \cup B = B \cup A$	$A \cap B = B \cap A$	commutative laws
$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$	absorption laws
$A \cup \varnothing = A$	$A \cap \Omega = A$	identity laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	distributive laws
$A \cup A^c = \Omega$	$A \cap A^c = \varnothing$	complement laws

Examples with algebra of events

Disjoint events

Events A and B are disjoint when they have no outcomes in common; in other words, if A happens, then B does not happen, and vice versa. In terms of the algebra of events, to say that A and B are disjoint is to say $A \cap B = \varnothing$.

“At least one”

Suppose we have events A, B, C. We can describe a new event informally by saying “at least one of A, B, C happens.” This event includes any outcome in A, B, or C. In terms of the algebra of events, the event “at least one of A, B, C happens” is $A \cup B \cup C$.

“Exactly one”

Suppose we have events A and B. We can describe a new event informally by saying “exactly one of A and B happens.” This event includes outcomes that are in A but not in B and outcomes that are in B but not in A. In terms of the algebra of events, the event “exactly one of A and B happens” is $(A \cap B^c) \cup (A^c \cap B)$. The events $A \cap B^c$ and $A^c \cap B$ are disjoint, since an outcome that is in both would be in A and in A^c, and those events are disjoint.

De Morgan’s laws

These are two useful formulas in the algebra of events relating $\cap$ and $\cup$ and complements:

- $(A \cap B)^c = A^c \cup B^c$

- $(A \cup B)^c = A^c \cap B^c$

There are similar formulas with more than two events. Algebra with numbers does not have any similar formulas.

Partitioning an event

Let A and B be any events. We then have $A = (A \cap B) \cup (A \cap B^c)$. You can reason out that equation by convincing yourself that every outcome in A is either in $A \cap B$ or $A \cap B^c$, or you can deduce it from the algebra of events as follows:

$\Omega = B \cup B^c$ complement law: every outcome is either in B or B^c
$A \cap \Omega = A \cap (B \cup B^c)$
$A \cap \Omega = (A \cap B) \cup (A \cap B^c)$ one of the distributive laws
$A = (A \cap B) \cup (A \cap B^c)$ identity law: $A \cap \Omega = A$

The events $A \cap B$ and $A \cap B^c$ are disjoint, since B and B^c are disjoint.